

RESEARCH ARTICLE

Robust control of vortex-induced vibration in flexible bridges using an active tuned mass damper

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Funding information

Program of Chang Jiang Scholars of Ministry of Education; Key Research and Development Plan of Anhui Province, Grant/Award Number: 202104g01020002; National Program on Key R&D Project of China, Grant/Award Number: 2020YFB2103500; Jiangsu Planned Projects for Postdoctoral Research Funds, Grant/Award Number: 2021K099A; China Postdoctoral Science Foundation, Grant/Award Number: 2021M690619; National Natural Science Foundation of China, Grant/Award Number: 52108442

Summary

The tuned mass damper (TMD) is widely used to mitigate vortex-induced vibration (VIV) that approximates harmonic vibration in bridges. Robust performance is, however, still a significant concern in the VIV mitigation of bridges, especially when using a TMD with a small mass ratio. This study presents a study to apply an active tuned mass damper (ATMD) to control VIV in bridges. The control algorithm is independent of bridge parameters and can achieve real-time tuning to the bridge motion by a few measurements. An adaptive TMD, obeying the parameter-independent algorithm, is developed to improve its robust performance when exposed to uncertainty in the natural frequency of the bridge, and then, the ATMD is used to implement such an adaptive TMD. The ATMD control using the LQG algorithm is used as a comparison to illustrate the advantage of the parameter-independent algorithm in the VIV mitigation of bridges. The effect of the control-structure interaction on the control effect is also discussed. Demonstration of the method on a numerical simulation of a long-span box-girder bridge confirms the potential superiority of the ATMD control using the parameter-independent algorithm in the VIV mitigation of bridges.

KEYWORDS

active tuned mass damper, adaptive tuned mass damper, control-structure interaction, robust control, vortex-induced vibration

1 | INTRODUCTION

Vortex-induced vibration (VIV), a common form of wind-induced vibration in long-span bridges, occurs in flexible bridges at moderate wind velocities.^{1,2} VIV is primarily a resonant vibration due to periodic excitations generated by vortex shedding behind a bluff body.³ A large-amplitude oscillation is locked within a range of shedding frequencies that bracket the natural frequency of the bluff body. This behavior is known as lock-in phenomena.⁴ Although the VIV of bridges at lock-in is not as serious as flutter, it impacts the fatigue life and serviceability of bridges.⁵

Since VIV of bridges is prevalent because it occurs at a moderate wind velocity, the control of VIV is necessary in practice.³ Generally, two mitigation strategies are options for the VIV mitigation, aerodynamic countermeasures and mass-damper control devices.^{6,7} The former strategy can be achieved by installing or optimizing fairings, flaps, guide vanes, or deflectors.^{6–8} However, these aerodynamic countermeasures may actually result in either beneficial or adverse effect on bridges having different sections.^{5,6} The latter strategy mainly adopts the tuned mass damper (TMD) to mitigate VIV in bridges, such as the Great East Belt Bridge,⁹ Trans-Tokyo Bay Bridge,⁵ Rio-Niterói Bridge,⁴ Jindo Bridge,¹⁰ and Hong Kong-Zhuhai-Macao Bridge.¹¹ In addition, the active tuned mass damper (ATMD),⁴ pounding TMD,¹² semi-active TMD,¹³ and TMD inerter^{14,15} are also being considered to mitigate VIV in bridges.

Certain limitations also exist currently in the applicability of TMDs for VIV control, including poor robustness to mistuning, limited frequency band control, and stroke limitation.¹⁶ It is challenging to predict the natural frequency of a bridge before it is constructed and in service. Because VIV exhibits narrow-frequency-band and low-frequency characteristics, robust performance is a major goal for using TMDs to control VIV in bridges.¹⁷ To improve the robust performance, robust design,^{18,19} ATMD,^{4,20–22} and semi-active TMD^{23–25} have been considered in recent decades. Although robust design can improve the robust performance to uncertainties in the primary structure, the improvement is not significant, especially in a case with a small mass ratio and large uncertainties.¹⁸ Compared with passive strategies, active and semi-active strategies attract more attention in TMD control. Certain semi-active TMDs may be viewed as aiming for real-time tuning based on measurements, such as the adaptive stiffness TMD,²⁶ magnetorheological TMD,²⁴ and eddy current TMD.²⁵ Nevertheless, in cases of large uncertainty, the semi-active TMD control effect does have limits due to the nature of semi-active devices. ATMDs are not limited in this way and thus do have potential benefits in performance, which are balanced by the greater amount of power needed to drive the auxiliary mass.¹⁶ The performance of an ATMD greatly depends on its control strategy and control algorithm. Among the variety of control algorithms available, the linear quadratic regulator (LQR)⁴ and linear quadratic Gaussian (LQG)²⁷ are widely used for ATMD systems. Note that these popular control algorithms are established on the basis of a linear time-invariant system. However, strictly speaking, the real structure is a nonlinear time-varying system due to the variability and uncertainty in structures and varying external disturbances. As a result, adaptive control algorithms have been proposed with the goal to maintain high efficiency in the ATMD system, such as fuzzy logic control,²⁸ wavelet particle swarm optimization-based LQR,²⁹ and multiscale wavelet-LQR.³⁰ Generally, each adaptive control algorithm deals with a specific problem in ATMD control in isolation, such as nonlinearity in structure,²⁸ control weights adjustment,²⁹ and instantaneous optimal control.³⁰

Just a few studies have focused on the robust control of VIV in bridges using mass dampers. In mitigating VIV in bridges, the mass ratio of the TMD is always small, less than 2%, due to realistic considerations including the installation space and carrying capacity. Thus, robustness in the TMD design is more consequential.¹¹ Passive strategies including off-line adjustment and on-line monitoring are costly, while an ATMD with adaptation may be a better option. In this study, the authors develop an ATMD approach based on an adaptive control algorithm that is independent of bridge parameters. The uncertainty mentioned here is the uncertainty in the natural frequency of the bridge, i.e., the shedding frequency within the lock-in region. In this approach, the dynamic coupling between the ATMD (including the auxiliary mass and hydraulic actuator) and bridge, or the control-structure interaction (CSI),³¹ is considered in our approach because unmodeled or improperly modeled CSI may severely affect the stability and control effectiveness.^{31,32}

In this paper, a parameter-independent algorithm is proposed to improve the control robustness of the ATMD in the VIV mitigation of bridges. The approach incorporates the kinematic Kalman filter to achieve real-time tuning by only measuring the stroke of the ATMD, the acceleration of the bridge, and the acceleration of the ATMD. The LQG control is also used as a comparison to illustrate the advantage of the parameter-independent algorithm. The effect of CSI on the control performance is also discussed in a well-tuned case and in the mistuning case. A long-span, box-girder bridge is used as numerical example to demonstrate the method and confirm the superiority of this parameter-independent algorithm in the VIV mitigation of bridges with ATMDs.

2 | VORTEX-INDUCED VIBRATION IN BRIDGES

The equation of motion of a bridge subjected to the VIV can be expressed as

$$\bar{m}(x) \frac{\partial^2 y(x,t)}{\partial^2 t} + c(x) \frac{\partial y(x,t)}{\partial t} + EI(x) \frac{\partial^4 y(x,t)}{\partial^4 x} = f_y(x,t), \quad (1)$$

where y is the vortex-induced amplitude of the bridge deck; $\bar{m}$ is the mass per unit length; c and EI are the damping coefficient and rigidity of the bridge; f_y is the vortex-induced forced acting on the bridge; and x represents the position of the bridge section.

In our study, the nonlinear semi-empirical model proposed by Ehsan and Scanlan¹ is used to describe the vortex-induced force

$$f_y(x,t) = \frac{1}{2} \rho U^2 D \left[Y_1(K^*) \left(1 - \varepsilon \frac{y^2}{D^2} \right) \frac{\dot{y}}{U} + Y_2(K) \frac{y}{D} + C_L(K^*) \cos(\omega t + \theta) \right], \quad (2)$$

where ρ is the air density; U is the mean wind velocity; D is the cross-wind dimension of the bridge; K^* is the reduced vortex shedding frequency; ω is the vortex shedding frequency; Y_1 , ε , and Y_2 aerodynamic parameters obtained from wind tunnel tests; C_L are the lift coefficients; and θ is the initial phase.

Since VIV in bridges is often dominated by a single mode, either a vertical mode or a torsional mode,^{1,17} vortex-induced amplitude of the bridge can be expressed by $y(x,t) = \phi(x)\eta(t)$. Premultiplying $\phi^T(x)$ and integrating from 0 to L at each side of Equation (1),

$$\int_0^L \phi^T(x) \bar{m}(x) \phi(x) dx \ddot{\eta}(t) + \int_0^L \phi^T(x) c(x) \phi(x) dx \dot{\eta}(t) + \int_0^L \phi^{''T}(x) EI(x) \phi''(x) dx \eta(t) = \int_0^L \phi^T(x) f_y(x,t) dx, \quad (3)$$

where $\phi(x)$ is either a vertical or a torsional mode of the bridge and L is the length of the bridge.

When the vortex shedding frequency approaches a modal frequency of the bridge, VIV with a large amplitude will be locked in at a wind velocity range. Within this lock-in region, the second and third terms of the right side of Equation (2), which respectively represent the frequency shift feature and forced excitation, can be dropped.^{1,4,17} Furthermore, by assuming full span-wise correction of the vortex-induced force,¹ Equation (3) can be expressed as

$$\ddot{\eta}(t) + 2\xi_{b,0}\omega_b\dot{\eta}(t) + \omega_b^2\eta(t) = \frac{1}{2M}\rho U^2 D Y_1 \left(\alpha_2 - \alpha_4 \varepsilon \frac{\eta^2}{D^2} \right) \frac{\dot{\eta}}{U}, \quad (4)$$

where $\xi_{b,0}$ and ω_b are the modal damping ratio and modal frequency of the bridge, respectively; $\alpha_n = \int_0^L \phi^n(x) dx$ ($n=2,4$). Note that Equation (4) can be rewritten in the form of the Van der Pol oscillator,^{17,33} a classical nonlinear oscillator, which is widely used to describe such self-excited and self-limited vibration, as

$$\ddot{\eta} + (\nu\eta^2 + 2\xi_b\omega_b)\dot{\eta} + \omega_b^2\eta = 0, \quad (5)$$

where $2\xi_b\omega_b = 2\xi_{b,0}\omega_b - \alpha_2\rho U D Y_1/(2M)$ and $\nu = \alpha_4\varepsilon\rho U Y_1/(2MD)$. Within the lock-in region, ξ_b can be considered as a negative damping ratio of the bridge.

3 | PASSIVE TMD CONTROL

3.1 | Bridge-TMD system

Consider a TMD is installed in the bridge with the section position x_t , as shown in Figure 1. The equation of motion of the TMD can be expressed as

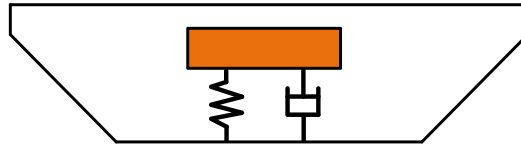


FIGURE 1 The bridge-TMD system

TABLE 1 Design formulas of TMD parameters

Passive TMD	Free vibration ^{34,35} (TMD-1)	Harmonic force ³⁶ (TMD-2)	Self-excitation ^{34,35} (TMD-3)
Frequency ratio	$\beta = \frac{1}{1+\mu}$	$\beta = \frac{1}{1+\mu}$	$\beta = \frac{1}{\sqrt{1+\mu}}$
Damping ratio of TMD	$\xi_t = \sqrt{\frac{\mu}{1+\mu}}$	$\xi_t = \sqrt{\frac{3\mu}{8(1+\mu)}}$	$\xi_t = \sqrt{\frac{\sqrt{1+\mu}-1}{2\sqrt{1+\mu}}}$

$$\phi(x_t)\ddot{\eta}_t(t) + 2\xi_t\omega_t(\phi(x_t)\dot{\eta}_t(t) - \phi(x_t)\dot{\eta}(t)) + \omega_t^2(\phi(x_t)\eta_t(t) - \phi(x_t)\eta(t)) = 0, \quad (6)$$

where ξ_t and ω_t are the damping ratio and natural frequency of the TMD, respectively. $\phi(x_t)\eta_t$ denotes the displacement of the TMD.

The TMD force acting on the bridge is written as

$$F_t(t) = \phi(x_t)^2 m [2\xi_t\omega_t(\dot{\eta}_t - \dot{\eta}(t)) + \omega_t^2(\eta_t - \eta(t))]. \quad (7)$$

Introducing the mass ratio $\mu = \phi(x_t)^2 m/M$ and frequency ratio $\beta = \omega_t/\omega_b$, the equations of motion of the bridge and TMD can be expressed as

$$\ddot{\eta} + (\nu\eta^2 + 2\xi_b\omega_b)\dot{\eta} + \omega_b^2\eta = \mu[2\beta\xi_t\omega_b(\dot{\eta}_t - \dot{\eta}) + \beta^2\omega_b^2(\eta_t - \eta)], \quad (8)$$

$$\ddot{\eta}_t + 2\beta\xi_t\omega_b(\dot{\eta}_t - \dot{\eta}) + \beta^2\omega_b^2(\eta_t - \eta) = 0. \quad (9)$$

This nonlinear system including the bridge and TMD can be represented in a general form as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \quad (10)$$

where $\mathbf{x} = \{\eta \ \dot{\eta} \ \eta_t \ \dot{\eta}_t\}^T$ is the state of the system.

3.2 | Robustness of passive TMD control

Here, passive TMDs with three types of design formulas are used to illustrate the poor robustness of the passive TMDs with a small mass ratio against the mistuning phenomenon and VIV uncertainty. These design formulas are presented in Table 1.

In the free-vibration TMD,^{34,35} the control objective is to maximize the minimum logarithmic decrement of the response, which is defined as $-2\pi\text{real}(\lambda)/\text{imag}(\lambda)$, where λ is the complex eigenvalue of the structure-TMD system. In the harmonic-force TMD,³⁶ the control objective is to minimize the peak of the structural displacement curve in the frequency domain when the structure subjected to the harmonic force. In the self-excitation TMD,^{34,35} the control objective is to make the real parts of two pairs λ be both zero.

Figure 2 shows the optimal β and ξ_t as calculated according to each of the three types of design formulas. Because the mass ratio is always small in VIV mitigation in bridges, differences in the optimal β will be small and can be neglected in the engineering design. However, there is considerable difference in the optimal ξ_t ; the free-vibration TMD and self-excitation TMD, respectively, yield designs with the largest and smallest damping ratios.

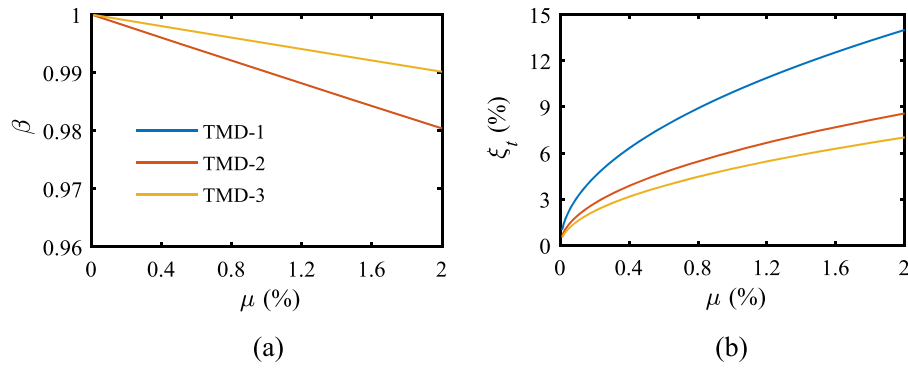


FIGURE 2 The optimal parameters of TMDs: (a) β and (b) ξ_t

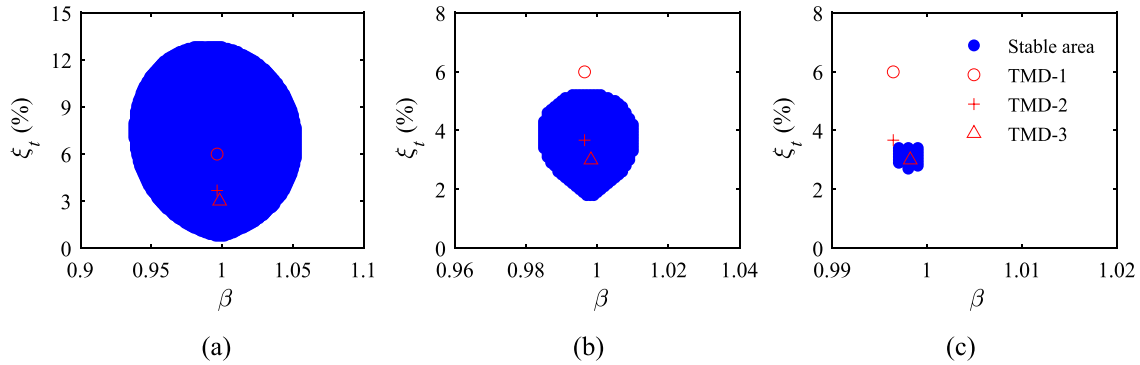


FIGURE 3 The region in which the TMD is stable for ξ_b : (a) $\xi_b = -0.7\%$, (b) $\xi_b = -1.7\%$, and (c) $\xi_b = -2.6\%$

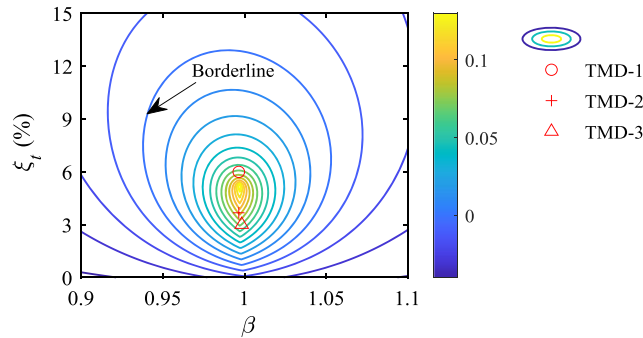


FIGURE 4 The contour map of the minimum damping ratio of the system

Figure 3 shows the region in which the TMD is stable for different ξ_b . The stable region is defined as the region corresponding to when the TMD parameters satisfies $\text{real}(\lambda_i) < 0$. The mass ratio is taken as 0.36%, a design value taken from a real bridge.¹¹ It can be concluded from this figure that the self-excitation TMD is the most robust to the uncertainty in a negative damping ratio of the bridge, i.e., the VIV uncertainty. Nevertheless, note that the change of ξ_b in a real bridge would be unlikely to be as significant as that shown in Figure 3.

Figure 4 shows the contour map of the minimum damping ratio of the bridge-TMD system. The region within borderline shown in Figure 4 refers to the stable region shown in Figure 3a. The harmonic-force TMD yields the largest minimum damping ratio, indicating the best control effectiveness. The free-vibration TMD and self-excitation TMD produce similar minimum damping ratios. Additionally, the free-vibration TMD generates a larger region of possible β to maintain system stability, followed by the harmonic-force TMD and self-excitation TMD. Uncertainty in β is common in real engineering, and it is significant for the TMD control with small mass ratio. Therefore, the free-vibration-type formulas are considered to be a robust choice for the parameter design of the TMD.

4 | ATMD CONTROL

4.1 | Bridge-ATMD system

As shown in Figure 4, the performance of the bridge-TMD system is very sensitive to the uncertainty in β even though the robust design formulas are used. In order to improve the robust performance of the TMD, the ATMD scheme is implemented by adding a hydraulic actuator between the mass and the bridge girder, as shown in Figure 5.

The dynamics of the hydraulic actuator and the attached structure is coupled by a natural velocity feedback link, i.e., CSI. Proper modeling of the actuator and the associated CSI can avoid producing unstable controllers and poor control performance.^{31,32} Maghareh et al.³² developed a model to describe the dynamics of the servo-hydraulic actuator and the CSI,

$$\dot{f}_a = -\alpha_2 \dot{x} - \alpha_3 f_a + r, \quad (11)$$

$$\dot{r} = -\alpha_1 \delta_0 x - \beta_1 r + \alpha_1 \delta_0 u, \quad (12)$$

where u denotes the control command; r denotes the dummy parameter; β_1 , $\alpha_1 \beta_0$, α_2 , and α_3 are the model parameters; and x and $\dot{x}$ denote the relative displacement and velocity of the hydraulic actuator, $x = (\eta_t - \eta)$ in the ATMD system. Figure 6 shows the block diagram for the dynamics of the servo-hydraulic actuator and CSI.

This nonlinear system including the bridge and ATMD can be represented by a general form,

$$\dot{\mathbf{x}} = \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{Bmatrix} = \mathbf{f}(\mathbf{x}, u, t) = \begin{Bmatrix} x_2 \\ -2\xi_b \omega_b x_2 - \nu x_1^2 x_2 - \omega_b^2 x_1 + \mu 2\beta \xi_t \omega_b (x_4 - x_2) + \mu \beta^2 \omega_b^2 (x_3 - x_1) - \frac{x_5}{M} \\ x_4 \\ -2\beta \xi_t \omega_b (x_4 - x_2) - \beta^2 \omega_b^2 (x_3 - x_1) + \frac{x_5}{m} \\ -\alpha_2 (x_4 - x_2) - \alpha_3 x_5 + x_6 \\ -\alpha_1 \delta_0 (x_3 - x_1) - \beta_1 x_6 + \alpha_1 \delta_0 u \end{Bmatrix}. \quad (13)$$

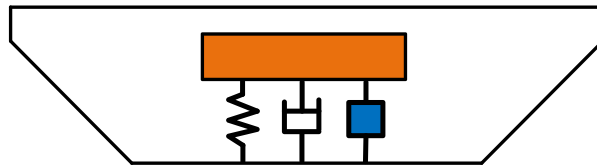


FIGURE 5 The bridge-ATMD system

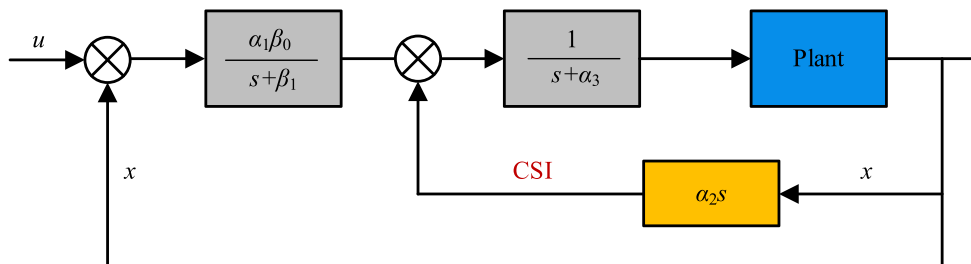


FIGURE 6 The block diagram for the dynamics of the servo-hydraulic actuator and CSI

4.2 | Parameter-independent algorithm

The parameter-independent algorithm is formulated based on an undamped dynamic vibration absorber (DVA).³⁷ Figure 7 shows the undamped structure equipped with an undamped DVA.

The equations of motion of the undamped DVA-structure system can be expressed as

$$M\ddot{X}_1 + KX_1 = F\sin\omega t + k(X_2 - X_1), \quad (14)$$

$$m\ddot{X}_2 + k(X_2 - X_1) = 0, \quad (15)$$

where X_1 and X_2 are the displacement responses of the structure and TMD relative to the ground; M and m are the masses of the structure and TMD; K and k are the stiffness coefficients of the structure and TMD; and F and ω are the amplitude and frequency of the harmonic force applied to the structure, respectively.

If the natural frequency of the undamped DVA is equal to the vibration frequency of the structure, and the external excitation is harmonic force, the vibration of the structure can be mitigated completely. Herein, this ideal undamped DVA is called adaptive TMD according to Frahm, and the associated control law is presented in Table 2. However, since there is no viscous element to dissipate the vibratory energy, the displacement response (or stroke) of the TMD is large, and the robustness of the system against the frequency control error is poor. In order to limit the stroke of the TMD and improve the robustness/stability, Weber and Mašlanka²⁴ added a damping force term into the desired force provided by the adaptive TMD, and the associated control law is presented in Table 2. The damping ratio is also adjusted in real-time based on the measured stroke of the TMD by changing the gain κ . A similar control algorithm was proposed by Sun and Nagarajaiah,²⁶ in which the damping ratio of the adaptive TMD adopted a zero-nonzero switch strategy. Note that the control law for the natural frequency of the adaptive TMD, $\bar{\omega}_2 = \omega$, is no longer optimal when the damping ratio of the adaptive TMD is introduced.

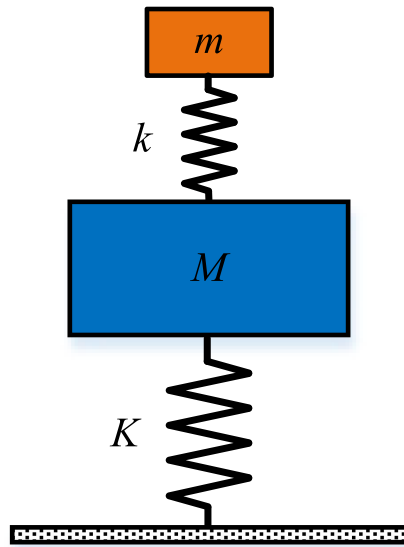


FIGURE 7 The undamped structure equipped with an undamped DVA

TABLE 2 Control algorithms used for adaptive TMDs

Adaptive TMD parameter	Adaptive TMD according to Frahm ³⁷	Weber's adaptive TMD
Natural frequency	$\bar{\omega}_{2,F} = \omega$	$\bar{\omega}_{2,W} = \omega$
Damping ratio	$\bar{\xi}_{2,F} = 0$	$\bar{\xi}_{2,W} = \kappa \sqrt{\frac{3\mu}{8(1+\mu)}}$

Before introducing our parameter-independent algorithm, the equations of motion of the adaptive TMD system are written as

$$M\ddot{X}_1 + KX_1 = F\sin\omega t + \bar{k}(X_2 - X) + \bar{c}(\dot{X}_2 - \dot{X})\#, \quad (16)$$

$$m\ddot{X}_2 + \bar{k}(X_2 - X) + \bar{c}(\dot{X}_2 - \dot{X}) = 0\#, \quad (17)$$

where $\bar{k}$ and $\bar{c}$ are the stiffness coefficient and damping coefficient of the adaptive TMD, respectively. The displacement dynamic amplification factors (DAFs) of the structure and adaptive TMD are expressed as

$$\frac{X_1(j\omega)}{X_{st}} = \frac{\omega_1^2(-\omega^2 + \bar{\omega}_2^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega)}{(-\omega^2 + \omega_1^2)(-\omega^2 + \bar{\omega}_2^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega) - \mu(\bar{\omega}_2^2\omega^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega^3)}\#, \quad (18)$$

$$\frac{X_2(j\omega)}{X_{st}} = \frac{\omega_1^2(\bar{\omega}_2^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega)}{(-\omega^2 + \omega_1^2)(-\omega^2 + \bar{\omega}_2^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega) - \mu(\bar{\omega}_2^2\omega^2 + 2j\bar{\xi}_2\bar{\omega}_2\omega^3)}\#, \quad (19)$$

where $X_{st} = F/K$ is the static displacement of the structure and $\omega_1 = \sqrt{K/M}$ is the natural frequency of the structure; $\bar{\omega}_2 = \sqrt{\bar{k}/m}$ and $\bar{\xi}_2 = \bar{c}/(2m\bar{\omega}_2)$ are the natural frequency and damping ratio of the adaptive TMD, respectively.

Since a well-tuned TMD is effective enough for the VIV mitigation,^{4,9} and the available space for a TMD to move within bridges is very limited,^{4,9,14} our control algorithm considers the damping ratio of the adaptive TMD to be a constant (not adaptive), $\bar{\xi}_{2,R} = \xi_2$, chosen as such to maintain a stable stroke limitation at any excitation frequency.

The adaptive natural frequency is written as a general form

$$\bar{\omega}_{2,R} = P(\omega - \omega_1) + Q\omega_1\#, \quad (20)$$

where P is the gain on the frequency deviation and Q can be considered as the frequency ratio when $\omega = \omega_1$. Because the objective is to improve the robustness of the TMD control to the uncertainty in ω_1 , the adaptive TMD should act to remain in a well-tuned state when a mistuning phenomenon occurs. Based on this goal, the frequency ratios at the true resonant frequency both in the well-tuned-case and in a mistuning case are as equal as possible as

$$\frac{\bar{\omega}_{2,R}(\omega_1)}{\omega_1} = Q \iff \frac{\bar{\omega}_{2,R}(\rho\omega_1)}{\rho\omega_1} = \frac{P(\rho\omega_1 - \omega_1) + Q\omega_1}{\rho\omega_1}\#, \quad (21)$$

where $\rho = \omega_{1,t}/\omega_1$ is the mistuning factor and $\omega_{1,t}$ is the true natural frequency of the structure. According to Equation (25), we can obtain $P = Q$, so that Equation (20) is simplified as

$$\bar{\omega}_{2,R} = P\omega\#. \quad (22)$$

Substituting Equation (22) into Equation (18) yields

$$\frac{X_1(j\omega)}{X_{st}} = \frac{(-1 + P^2 + 2iP\xi_2)}{(-\gamma^2 + 1)(-1 + P^2 + 2iP\xi_2) - \mu(\gamma^2 P^2 + 2iP\gamma^2 \xi_2)}\#, \quad (23)$$

where $\gamma = \omega/\omega_1$. Defining a function $H = |X_1(j\omega)/X_{st}|^2$, then the optimal P with the minimum value of $H_{\max}$, which is the maximum value of H over all γ , can be obtained by solving two equations:

$$\frac{\partial H(P, \xi_2, \gamma, \mu)}{\partial \gamma} = 0 \Rightarrow \gamma_0(P, \xi_2, \mu)\#, \quad (24)$$

$$\frac{\partial H(P, \xi_2, \gamma_0, \mu)}{\partial P} = 0 \Rightarrow P_{opt}(\xi_2, \mu)\#. \quad (25)$$

Finally, the analytical solution of P_{opt} is given by

$$P_{opt}^2 = \frac{\mu - 4\xi_2^2\mu - 4\xi_2^2 + 2 + \sqrt{16\xi_2^4\mu^2 + 32\xi_2^4\mu + 16\xi_2^4 - 8\xi_2^2\mu^2 - 24\xi_2^2\mu - 16\xi_2^2 + \mu^2 + 16\mu + 16}}{6(\mu + 1)}. \quad (26)$$

Because the mass ratio used here is small ($\mu = 0.36\%$), the optimal P is very close to 1 over the whole range of ξ_2 , as shown in Figure 8a. Figure 8b shows the maximum displacement DAFs of the structure and mass (adaptive TMD) with

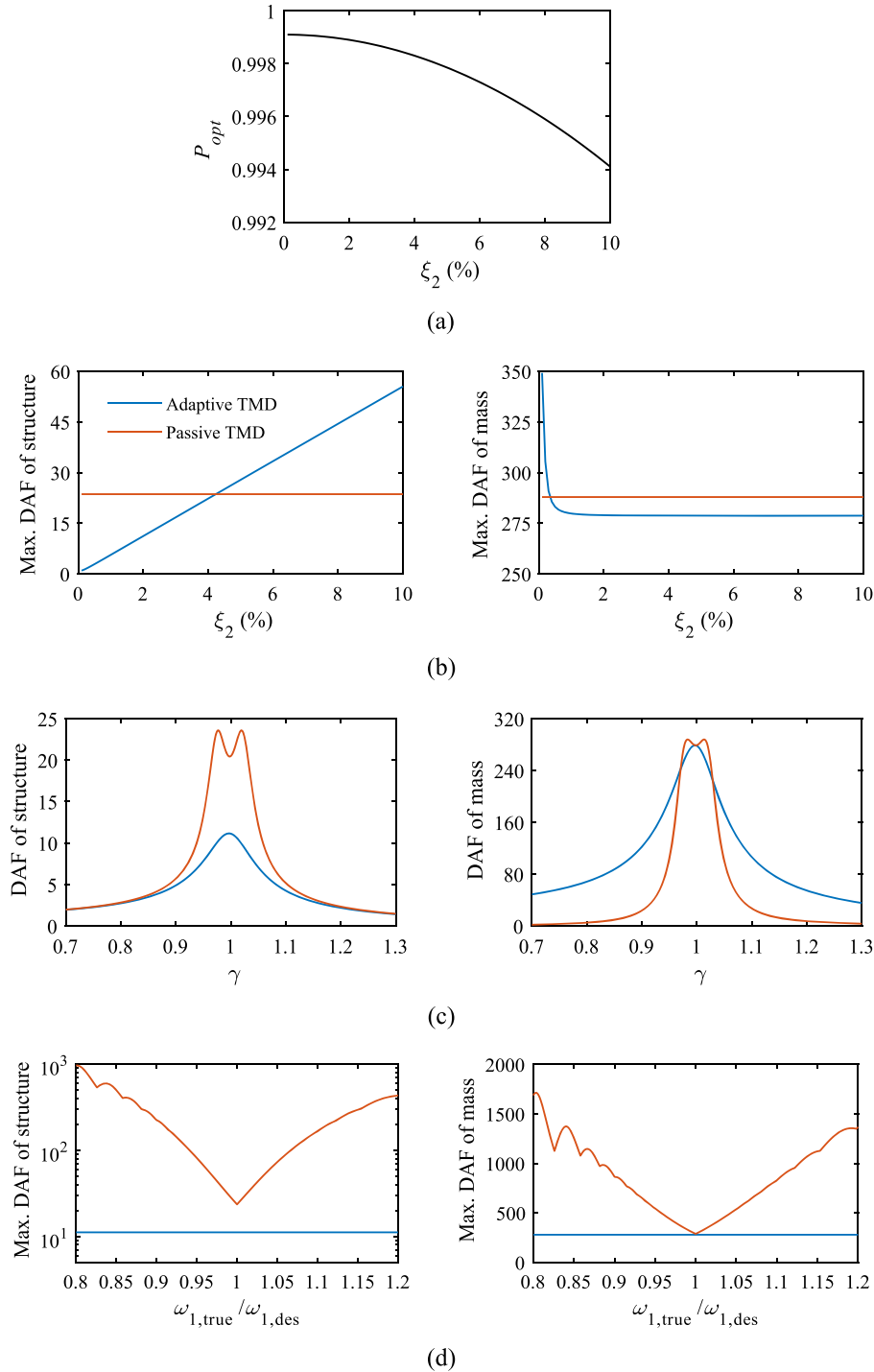


FIGURE 8 Comparison between passive TMD control and adaptive TMD control when $\mu = 0.36\%$: (a) the optimal P , (b) the maximum displacement DAFs, (c) the displacement DAF curves, and (d) the maximum displacement DAFs in the mistuning case

ξ_2 . The dotted line is the control result of the passive harmonic-force TMD with $\beta = 0.996$ and $\xi_2 = 3.67\%$. Within the range of small ξ_2 (0.42%), the adaptive TMD control is superior to the TMD control. For the stroke limitation, the adaptive TMD control gives a better result than the passive TMD control, except in the case of extremely small values of ξ_2 . Figure 8c shows the comparison of control results of the adaptive TMD with $\xi_2 = 2\%$ and the passive TMD. In addition to the superiority in the reduction of the maximum displacement DAF, the adaptive TMD control has a wider effective frequency range. Figure 8d shows the control results of the adaptive TMD with $\xi_2 = 2\%$ and the passive TMD in the mistuning case. Obviously, the adaptive TMD control is much more robust than the passive TMD control, and this advantage is more significant than that in the optimal control effect (i.e., well-tuned case) for VIV mitigation of bridges.

4.3 | ATMD control using parameter-independent algorithm

With the parameter-independent algorithm applied into the ATMD control, the total force provided by the adaptive TMD can be expressed as

$$f_{TMD} = m(P_{opt}\omega)^2(\eta_t - \eta) + 2m(P_{opt}\omega)\xi_t(\dot{\eta}_t - \dot{\eta}) = f_s - f_a \# \quad (27)$$

where $f_s = m\omega_t^2(\eta_t - \eta)$ is the elastic force provided by the stiffness element and f_a is the control force provided by the servo-hydraulic actuator. Here, the viscous damper is removed from the ATMD to reduce the control cost and uncertainty. The desired control force is given by

$$f_a = -m(P_{opt}\omega)^2(\eta_t - \eta) - 2m(P_{opt}\omega)\xi_t(\dot{\eta}_t - \dot{\eta}) + f_s \# \quad (28)$$

In the ATMD control, the control command u is the control input, rather than the control force f_a . The control command is obtained by solving the differential form of Equations (11) and (12):

$$f_a(nT + T) = e^{-\alpha_3 T} \cdot f_a(nT) - \int_0^T e^{-\alpha_3 t} \alpha_2 dt \cdot (\dot{\eta}_t - \dot{\eta})(nT) + \int_0^T e^{-\alpha_3 t} dt \cdot r(nT + T) \#, \quad (29)$$

$$r(nT + T) = e^{-\beta_1 T} \cdot r(nT) - \int_0^T e^{-\beta_1 t} \alpha_1 \delta_0 dt \cdot (\eta_t - \eta)(nT) + \int_0^T e^{-\beta_1 t} \alpha_1 \delta_0 dt \cdot u(nT + T) \#, \quad (30)$$

where T is the time step; $f_a(nT + T)$ denotes the desired control force at the next step; $f_a(nT)$ denotes the measured control force at the current step; and $u(nT + T)$ is the control command at the next step.

When calculating the control force and control command, these parameters ω , $(\eta_t - \eta)$, $(\dot{\eta}_t - \dot{\eta})$, f_s , and $f_a(nT)$ should be known. Considering that these parameters $(\eta_t - \eta)$, f_s , and $f_a(nT)$ can be measured by sensors, the determination of ω and $(\dot{\eta}_t - \dot{\eta})$ become the two issues in the ATMD control.

Since the VIV of the bridge during the lock-in region behaves self-excited characteristic, the excitation frequency ω can be considered as the dominant frequency of the bridge motion. Many time-frequency techniques can capture the dominant frequency, such as empirical mode decomposition, Hilbert transform, and short-term Fourier transform.³⁸ Unlike the multi-frequency component of earthquake records and multimodal participation of seismic structures,³⁸ the VIV in the bridge during the lock-in region is a single-mode resonance.^{1,17} As a result, it is relatively easy to identify its dominant frequency. Here, the zero-crossings of the absolute acceleration of the bridge are recorded in real time, as shown in Figure 9, and the dominant frequency is calculated by

$$\omega(k+1) = \frac{\pi}{(t_{k+1} - t_k)} \# \quad (31)$$

To reduce the effects of the measurement noise, a low-pass filter is used to deal with the measured acceleration of the bridge before identifying the dominant frequency

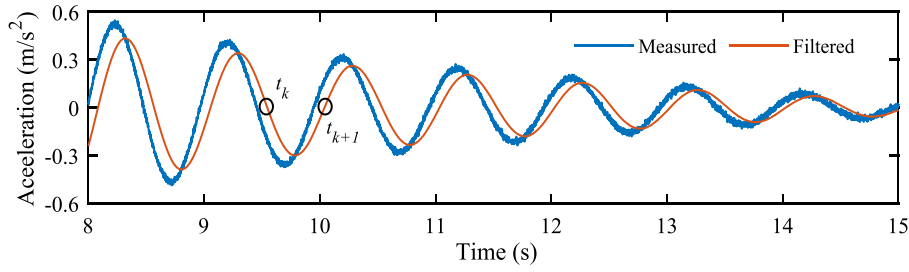


FIGURE 9 The measured acceleration and filtered acceleration

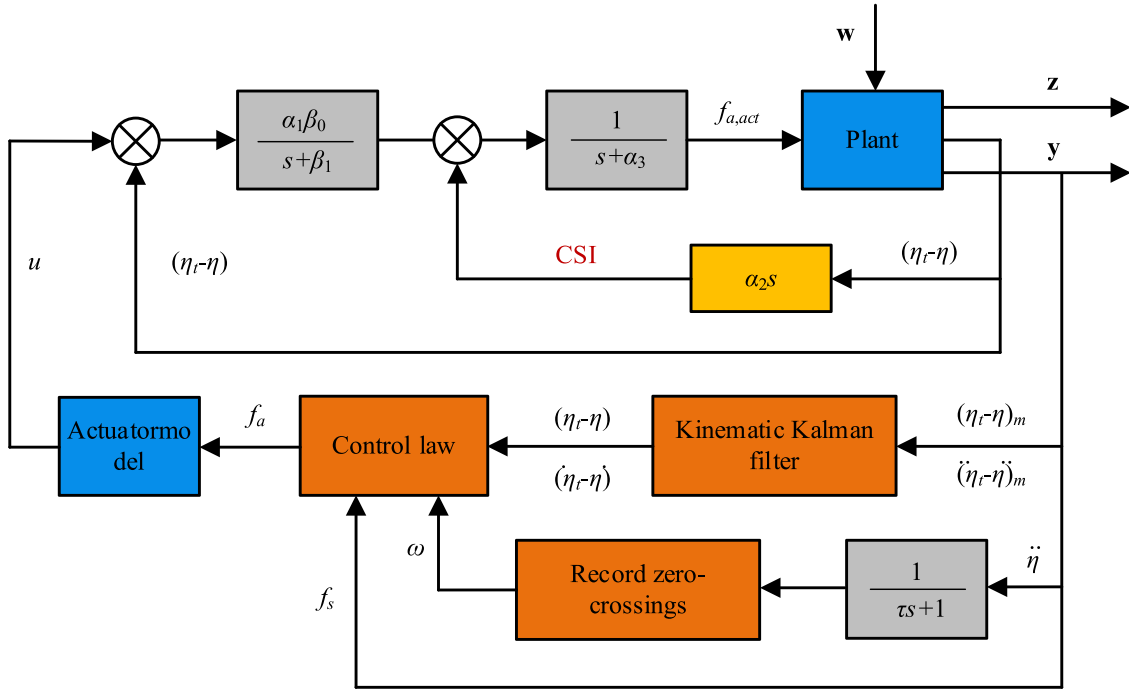


FIGURE 10 The block diagram of ATMD control system using the parameter-independent algorithm

$$H(s) = \frac{1}{s\tau + 1} \# , \quad (32)$$

where s is the Laplace transform variable and τ is the filter time constant. From Figure 10, the filtered acceleration curve is clearly much smoother than the measured acceleration of the bridge.

In model-based control algorithms, the Kalman filter is used to estimate the state vector of the system. Thus, the estimation accuracy of the state greatly depends on the model parameters. To improve the estimation accuracy of $(\dot{\eta}_t - \dot{\eta})$ in mistuning cases, a non-model-based Kalman filter, named kinematic Kalman filter,³⁹ is used here:

$$\begin{Bmatrix} \dot{x}_{1,K} \\ \dot{x}_{2,K} \end{Bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} x_{1,K} \\ x_{2,K} \end{Bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_K + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w_K \# , \quad (33)$$

$$y_K = [1 \ 0] \begin{Bmatrix} x_{1,K} \\ x_{2,K} \end{Bmatrix} + v_K \# , \quad (34)$$

where $x_{1,K}$ and $x_{2,K}$ represent the displacement and velocity, $(\eta_t - \eta)$ and $(\dot{\eta}_t - \dot{\eta})$; u_K represents the measured acceleration, $(\ddot{\eta}_t - \ddot{\eta})_m$; w_K is the acceleration noise; y_K represents the measured displacement, $(\eta_t - \eta)_m$; and v_K is the

displacement noise. From Equations (33) and (34), we find that the kinematic Kalman filter is established based on the kinematic relations between the displacement, velocity, and acceleration and it involves neither physical parameters nor external disturbances.³⁹ Besides, the kinematics Kalman filter is very well suited for Kalman filter theory. Figure 10 shows the block diagram of ATMD control system using the parameter-independent algorithm for the VIV mitigation of bridges.

4.4 | ATMD control using LQG algorithm

To illustrate the advantages of the parameter-independent algorithm for this application involving VIV control, the LQG algorithm is used here as a comparison. To apply LQG algorithm to the system, Equation (12) should be written as the linear form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}_1\mathbf{w} + \mathbf{B}_2u\#, \quad (35)$$

$$\text{where } \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\omega_b^2 - \mu\beta^2\omega_b^2 & -2\xi_b\omega_b - 2\mu\beta\xi_t\omega_b & \mu\beta^2\omega_b^2 & 2\mu\beta\xi_t\omega_b & -\frac{1}{M} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \beta^2\omega_b^2 & 2\beta\xi_t\omega_b & -\beta^2\omega_b^2 & -2\beta\xi_t\omega_b & \frac{1}{m} & 0 \\ 0 & \alpha_2 & 0 & -\alpha_2 & -\alpha_3 & 1 \\ \alpha_1\delta_0 & 0 & -\alpha_1\delta_0 & 0 & 0 & -\beta_1 \end{bmatrix}, \mathbf{B}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{B}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \alpha_1\delta_0 \end{bmatrix}.$$

The displacement of the bridge and the stroke of the ATMD are selected as system outputs, and the equation is given by

$$\mathbf{z} = \mathbf{C}_1\mathbf{x} + \mathbf{D}_{11}\mathbf{w} + \mathbf{D}_{12}u\#, \quad (36)$$

$$\text{where } \mathbf{C}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \mathbf{D}_{11} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ and } \mathbf{D}_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The absolute accelerations of the bridge and the ATMD and the stroke of the ATMD are selected as system measurements, and the equation is given by

$$\mathbf{y} = \mathbf{C}_2\mathbf{x} + \mathbf{D}_{21}\mathbf{w} + \mathbf{D}_{22}u + \mathbf{v}\#, \quad (37)$$

$$\text{where } \mathbf{C}_2 = \begin{bmatrix} -\omega_b^2 - \mu\beta^2\omega_b^2 & -2\xi_b\omega_b - 2\mu\beta\xi_t\omega_b & \mu\beta^2\omega_b^2 & 2\mu\beta\xi_t\omega_b & -\frac{1}{M} & 0 \\ \beta^2\omega_b^2 & 2\beta\xi_t\omega_b & -\beta^2\omega_b^2 & -2\beta\xi_t\omega_b & \frac{1}{m} & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \mathbf{D}_{21} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \text{ and } \mathbf{D}_{22} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}; \mathbf{v} \text{ denotes the measurement noise.}$$

Figure 11 shows the LQG control block diagram. The objective of LQG algorithm is to find an optimal gain matrix $\mathbf{K}$ minimizing the performance function $\mathcal{F}$:

$$\mathcal{F} = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} E \left[\int_0^\tau \left\{ (\mathbf{C}_1\mathbf{x})^T \mathbf{Q} (\mathbf{C}_1\mathbf{x}) + u^T \mathbf{R} u \right\} dt \right] \#, \quad (38)$$

where $E[\cdot]$ denotes expectation and $\mathbf{Q}$ and $\mathbf{R}$ are the weighting matrices of the output and the control command, respectively. The optimal control command is determined by adopting the control law:

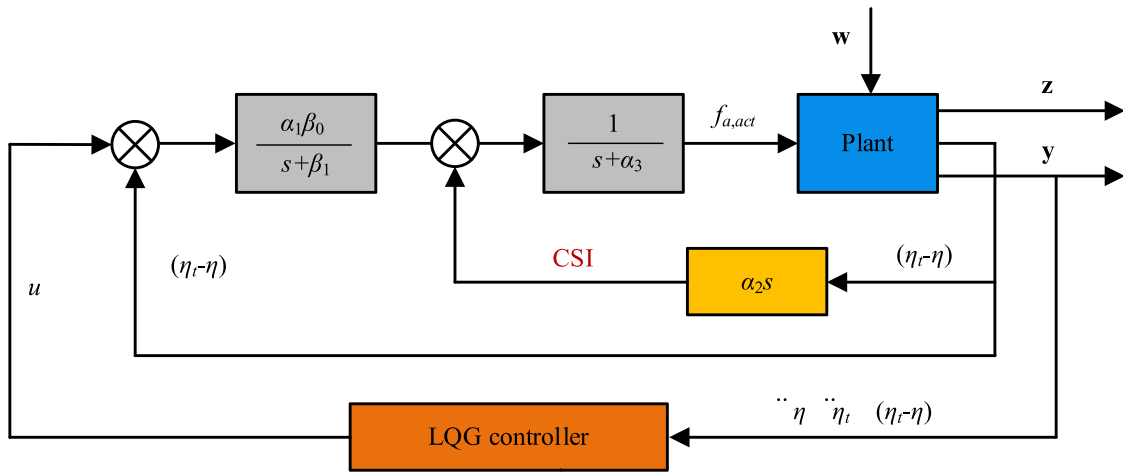


FIGURE 11 The LQG control block diagram

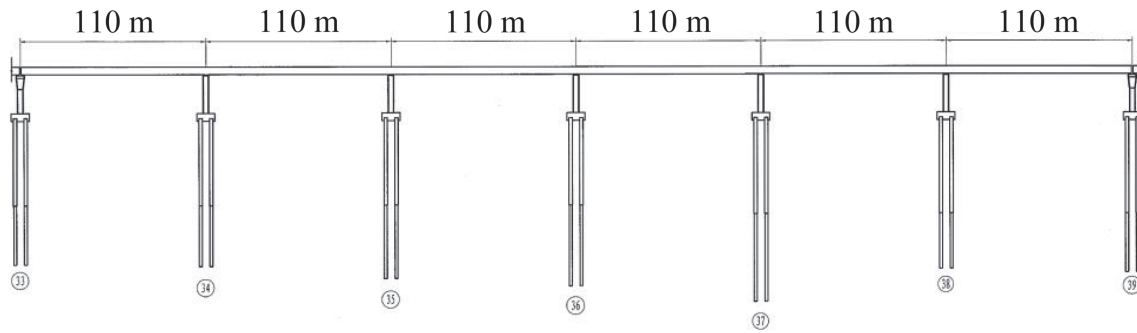


FIGURE 12 A standard link of the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge

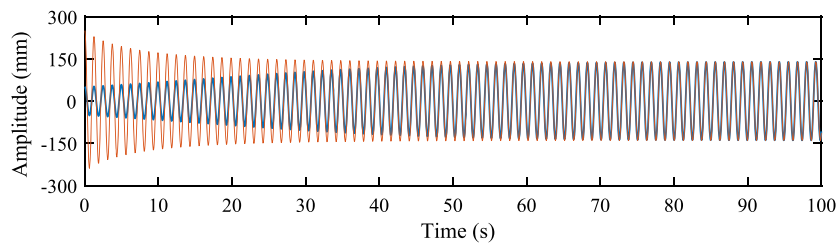


FIGURE 13 The vortex-induced amplitude of the bridge under two initial conditions

$$u = -\mathbf{K}\hat{\mathbf{x}}\#, \quad (39)$$

where $\hat{\mathbf{x}}$ is the estimated state vector from the Kalman filter.

5 | NUMERICAL EXAMPLE

5.1 | Bridge description

A numerical simulation of a flexible bridge is performed to validate the superiority of the ATMD control using the parameter-independent algorithm. The continuous steel box-girder bridge is a standard link of the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge, which is shown in Figure 12.

The ANSYS analysis showed that the natural frequency of first-order vertical bend mode of the bridge is 0.806 Hz, and the corresponding modal mass is 8888.9 tons.^{11,14} The damping ratio of the first-order vertical bending mode is assumed as 0.3%. Wind tunnel tests of the scaled sectional and full-bridge aeroelastic models show that the bridge will generate a large vortex-induced amplitude with 150 mm at the wind velocity of 27–35 m/s.¹¹ The fourth-order Runge-Kutta method is used to simulate the vortex-induced amplitude of the bridge within the lock-in region. Figure 13 shows the vortex-induced amplitude of the bridge under two initial conditions, $\eta(0) = 50$ mm and $\eta(0) = 250$ mm. It can be seen that the amplitude increases or decreases with time and then both keep stable value, 150 mm, which indicates that the VIV of the bridge is a self-excited and self-limited vibration.

5.2 | Control result of TMD

Since the 150 mm amplitude exceeds the limit value suggested by China code, 50 mm, the TMD scheme is used to mitigate the VIV of the bridge. The eight TMDs were installed in the center of the second span and fifth span, with a total mass of 32 tons, which provides a mass ratio of 0.36%. Three passive TMDs designed according to the design formulas presented in Table 1 are used here. The initial conditions in this simulation is $\eta(0) = 20$ mm. It can be seen from Figure 14a that the three passive TMDs all completely eliminate the vortex-induced amplitude of the bridge. Besides, the self-excitation TMD control has the shortest decay time, followed by harmonic-force and free-vibration TMD controls. As a result, the self-excitation TMD has the largest stroke, followed by harmonic-force and free-vibration TMD controls, as shown in Figure 14b.

Figure 15 shows the vortex-induced amplitude of the bridge with a mis-tuned TMD. The relative errors of the natural frequency of the bridge in Figure 15a,b are -5% and $+5\%$, respectively. Among three TMDs, the free-vibration TMD has the best mitigation effect on the VIV of the bridge, which confirms the result in Section 3.2. However, the three TMD all cannot completely mitigate the VIV of the bridge, and the result will be worse in the larger error case. In general, the control robustness of the TMD is not good even though the robust design formulas are used.

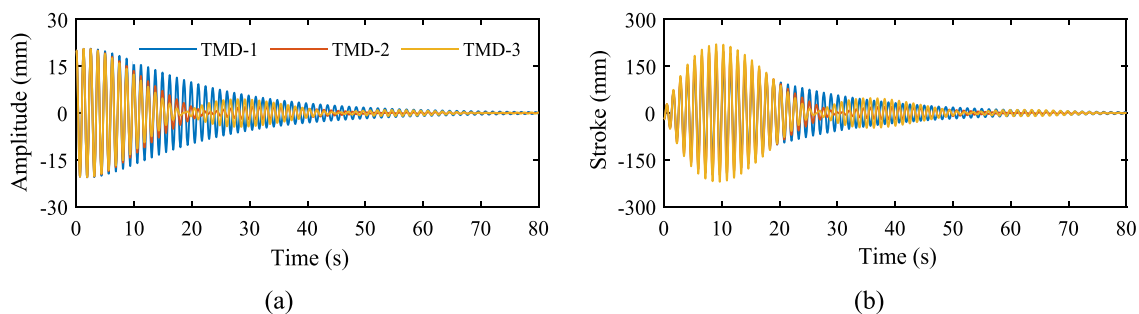


FIGURE 14 The displacement responses of the bridge-well-tuned TMD system: (a) the bridge and (b) the TMD

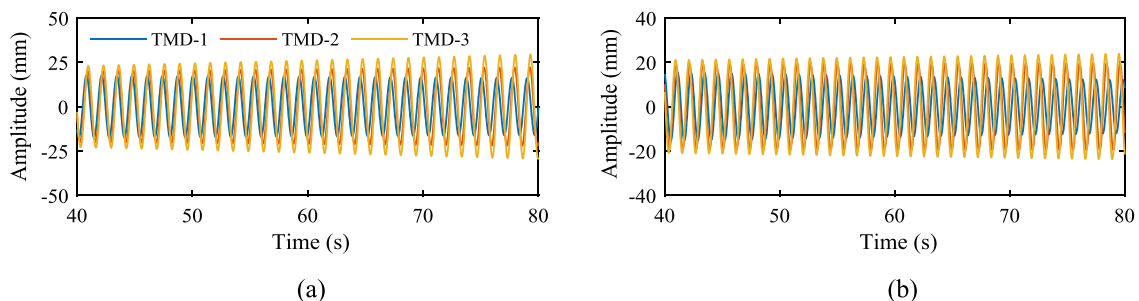


FIGURE 15 The vortex-induced amplitude of the bridge with a mis-tuned TMD: (a) -5% and (b) 5%

5.3 | Control result of ATMD

In ATMD control, whether the parameter-independent algorithm or the LQG algorithm is used, the mass ratio is 0.36%, and the stiffness of the spring is determined according to the free-vibration TMD. In the parameter-independent algorithm, ξ_t is designed as 5% to limit the stroke of the ATMD. According to Equation (26), the optimal frequency gain P_{opt} is designed as 0.9979. The time constant of the lowpass filter is adopted as 0.15 to effectively smooth the absolute acceleration. To reduce the effect of the identification error of ω in the case of the low acceleration signal, the upper and lower bounds of the identified ω are added in the algorithm. Considering that the natural frequency of the first vertical bending mode of the bridge is 0.806 Hz, the upper and lower bounds are designed as 1.1 and 0.5 Hz. In the LQG algorithm, the weighting matrices are designed as $\mathbf{Q} = \begin{bmatrix} 100 & 0 \\ 0 & 20 \end{bmatrix}$ and $\mathbf{R} = 1$ to keep the same maximum stroke as that in the parameter-independent algorithm. In all the simulations, the hydraulic actuator is in the turn-off state within the first 10 s; i.e., the control command is 0.

Figure 16 shows the controlled results with the LQG and parameter-independent algorithms. The zero-crossing-based method can effectively identify the vortex shedding frequency (or the natural frequency of the bridge) within the first 50 s. In the last 30 s, the identification accuracy is not sufficiently accurate due to the low-amplitude of the accelerations measured. But the error is not significant for the control effect, because the VIV is greatly mitigated and the desired control force is very small that can be neglected in the last 30 s. Note again that the vortex shedding frequency is estimated from the acceleration measurement of the bridge; since the vibration frequency of the bridge under VIV within the lock-in region is very close to the natural frequency of the bridge,^{1,17} the vortex shedding frequency is basically identical with the natural frequency of the bridge. In fact, the frequency estimation method used here is similar to those in references Weber and Mašlanka²⁴ and Weber.⁴⁰ In the control system with the turn-off state, the ATMD cannot play a role as the TMD to mitigate the VIV due to the existence of the CSI. As a result, the ATMD fails to mitigate the VIV within the first 10 s. Once the ATMD works, the vortex-induced amplitude decays rapidly with time, which indicates that the VIV is completely mitigated. Since the LQG control belongs the optimal control, the LQG algorithm gives a larger control force and a faster amplitude decay than the parameter-independent algorithm at the beginning of the control, between 10 and 24 s. Since the LQG algorithm chooses to simultaneously weight the amplitude reduction and the stroke limitation, the peak control force is larger than that in the parameter-independent algorithm. In the

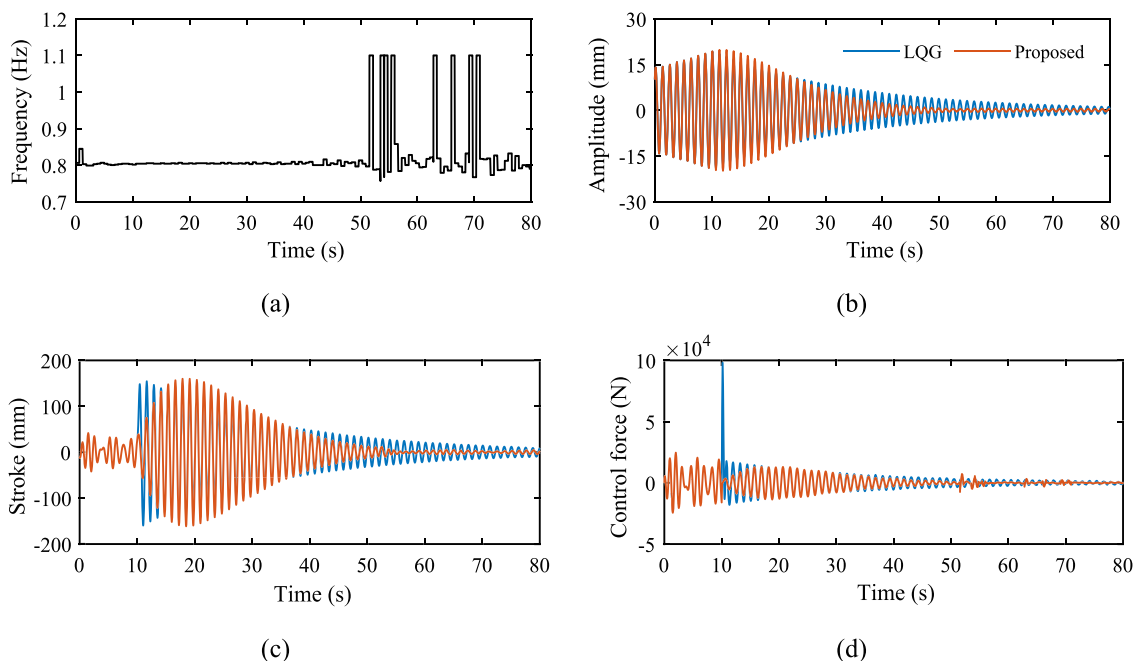


FIGURE 16 The controlled results with the LQG and parameter-independent algorithms: (a) the estimated frequency, (b) the amplitude of the bridge, (c) the stroke of the TMD, and (d) the control force

remaining time, the parameter-independent algorithm gives better control result in the amplitude reduction and the stroke limitation, because the LQG algorithm based on the white noise input assumption is focused on the vibration control over the whole range of frequencies, as a result, the provided control force is not optimal for the VIV compared to that in the parameter-independent algorithm. In general, on the basis of equal maximum strokes, the parameter-independent algorithm has better control performance than the LQG algorithm in the VIV mitigation of the bridge.

Next, we consider the mis-tuning case, in which the true natural frequency of the bridge is 80 % of the design value. Figure 17 shows the controlled results with the LQG and parameter-independent algorithms in the mis-tuning case. We find that the zero-crossing-based method can quickly identify the vortex shedding frequency, from 0.806 to 0.645 Hz. Since the implementation of the LQG algorithm is based on the initial design state, the control force calculated by the LQG algorithm is longer not optimal for the VIV mitigation. As a result, the ATMD with the LQG algorithm cannot eliminate the VIV in the limited time. However, the ATMD with the parameter-independent algorithm rapidly eliminates the VIV, indicating that the proposed algorithm is robust enough to the error in ω_b . Apparently, the parameter-independent algorithm uses a larger stroke and control force than the LQG algorithm in the crucial working time of the ATMD, between 15 and 35 s. From the comparison of two control forces, we find that the parameter-independent algorithm gives a force-stroke curve with a larger slope to better tune the frequency than the LQG algorithm.

In order to further compare the two control algorithms, some evaluation criteria are defined in terms of the maximum and root mean square (RMS) values, for responses including maximum amplitude, stroke, control force, and

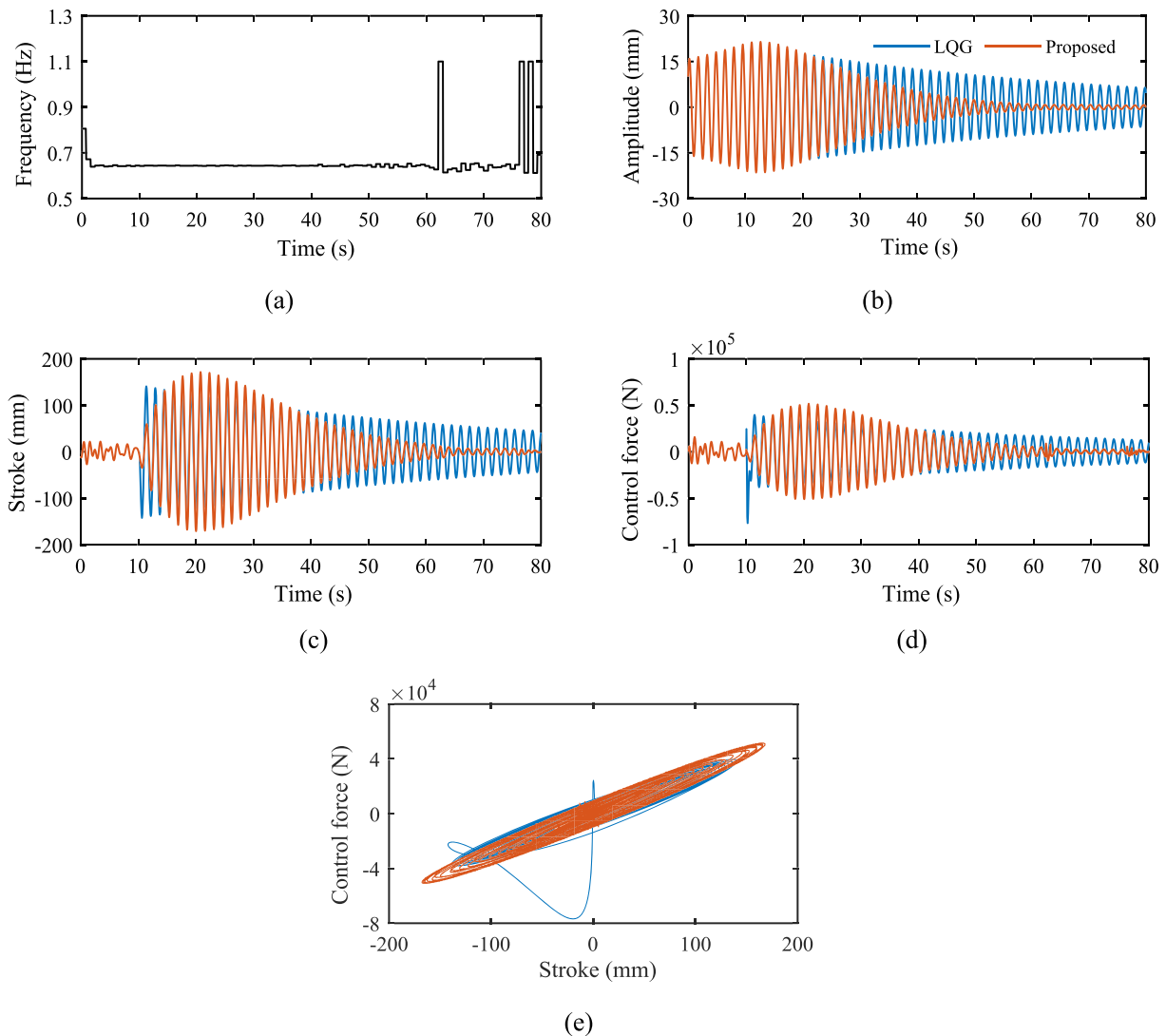


FIGURE 17 The controlled results with the LQG and parameter-independent algorithms in the mis-tuning case: (a) the estimated frequency, (b) the amplitude of the bridge, (c) the stroke of the TMD, (d) the control force, and (e) the control force-stroke curve

RMS amplitude, stroke, control force between 10 and 80 s. Figure 18 shows the maximum and RMS responses of the ATMD control with different errors in ω_b . Since the maximum amplitude occurs at the beginning of the control, and the LQG algorithm gives a large control force immediately, the LQG algorithm has better control effect than the parameter-independent algorithm in the maximum amplitude, as shown in Figure 18a. In fact, the RMS amplitude is a better criterion to evaluate the control effect in the VIV mitigation of bridges. It can be seen that the parameter-independent algorithm has much better control effect than the LQG algorithm when the error is -30% , -20% , and -10% . From the result in Figure 16b, we find that the LQG algorithm has better control effect at the beginning of the control and the corresponding amplitude is large. As a result, the RMS amplitude using the parameter-independent algorithm is slightly larger than that using LQG algorithm when the error is 10% and 20% . Note that the VIV of the bridge has been eliminated by the ATMD with the parameter-independent algorithm when the error is 10% and 20% . When the error is -30% and -20% , the stroke of the auxiliary mass using the parameter-independent algorithm is larger than that using LQG algorithm, which indicates that the LQG algorithm cannot achieve a tuning effect in the large error in ω_b . Due to sufficient damping ratio in the parameter-independent algorithm, $\xi_t = 5\%$, the strokes of the auxiliary mass in all the mis-tuning cases are not large and can be accepted in practice. Although the LQG algorithm

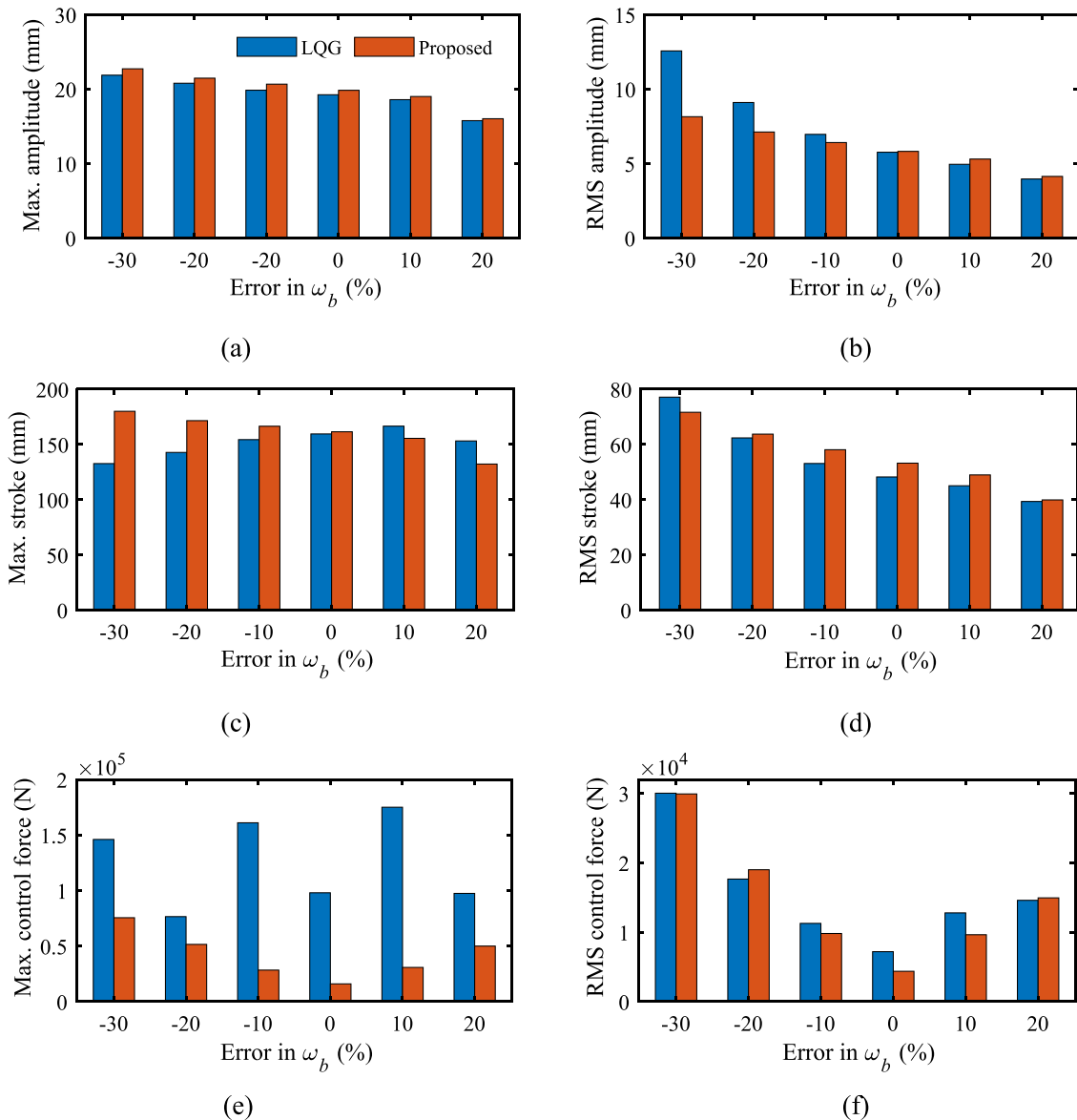


FIGURE 18 The maximum and RMS responses of the ATMD control with different errors in ω_b : (a) the max. amplitude, (b) the RMS amplitude, (c) the max. stroke, (d) the RMS stroke, (e) the max. control force, and (f) the RMS control force

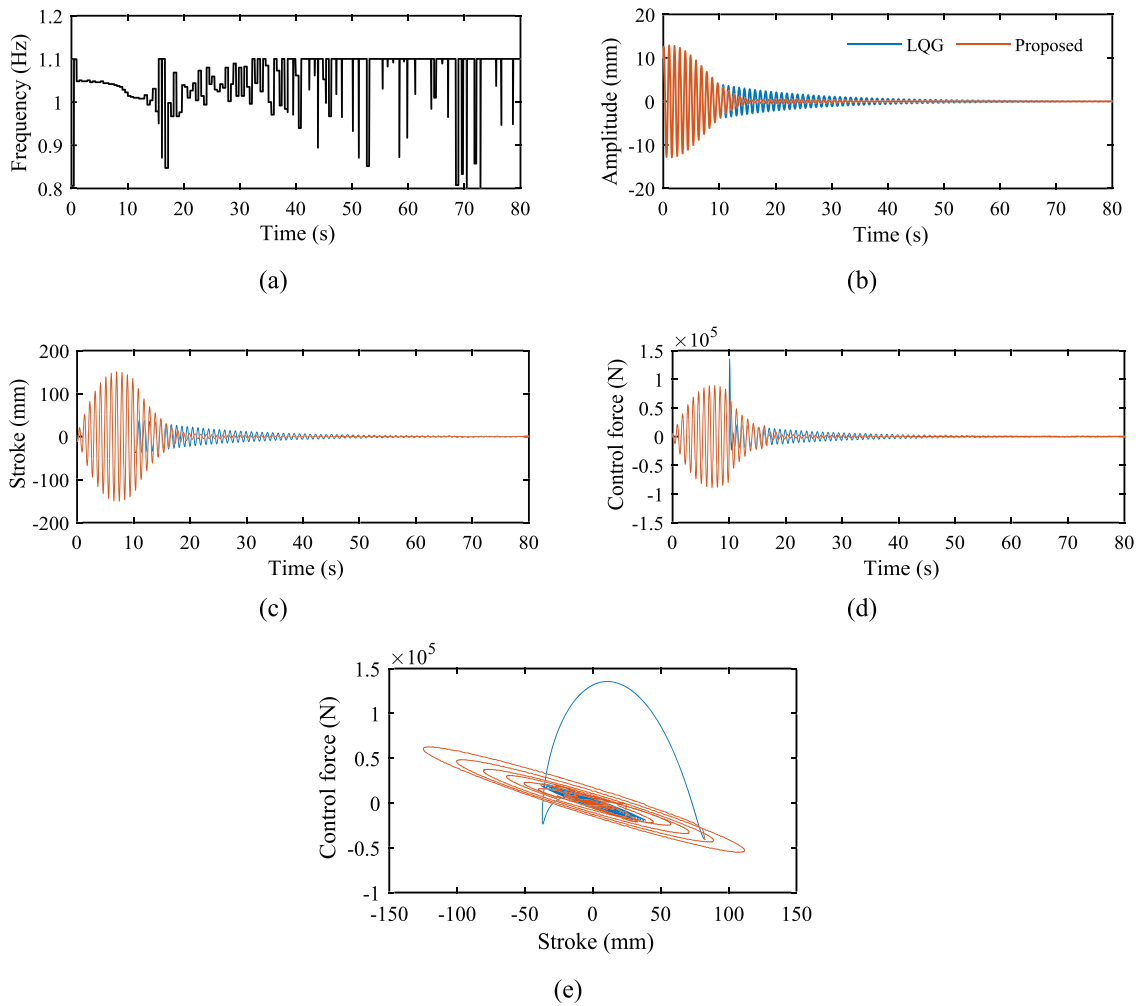


FIGURE 19 The controlled results with the LQG and parameter-independent algorithms: (a) the estimated frequency, (b) the amplitude of the bridge, (c) the stroke of the TMD, (d) the control force, and (e) the control force-stroke curve

has better instantaneous control, it results in a larger instantaneous control force than the parameter-independent algorithm, which greatly increases the capacity requirement of the servo-hydraulic actuator.

Finally, we consider a special mis-tuning case; the true natural frequency of the bridge is 130% of the design value. Figure 19 shows the controlled results with the LQG and parameter-independent algorithms. It can be seen that the zero-crossing-based method can identify the true natural frequency of the bridge only in the first 15 s, because the VIV has been basically mitigated by the ATMD before the ATMD works, and in the remaining time, the noise in the small-amplitude acceleration measurement greatly reduces the identification accuracy. Unlike other mis-tuning cases, in this case the ATMD with the turn-off state play a role as a well-tuned TMD, as a result, the VIV of the bridge is greatly mitigated in the first 10 s. The reason for this is that the CSI presented in Equations (11) and (12) is equivalent to a stiffness-like element between the auxiliary mass and the bridge when control command is zero. This special stiffness plus the stiffness provided by the spring make the natural frequency of the mass be tuned to the true natural frequency of the bridge, 1.048 Hz.

6 | CONCLUSION

Unlike the commonly robust control,⁴¹ this paper proposed a robust control of the VIV that approximates harmonic vibration in flexible bridges using an ATMD to implement an adaptive TMD obeying a parameter-independent algorithm. The parameter-independent algorithm used, which incorporated the kinematic Kalman filter, was independent of bridge parameters, and it was implemented by only measuring the stroke of the ATMD and accelerations of the

bridge and ATMD. The advantage of the ATMD control using the parameter-independent algorithm has been demonstrated through a numerical example using a full-scale bridge model. The numerical results showed that, under the premise of equal maximum strokes, the parameter-independent algorithm gave a similar control result as the LQG algorithm in the vortex-induced amplitude reduction of the bridge while it required a much smaller maximum control force. In various mistuning cases, the parameter-independent algorithm was more robust than the LQG algorithm, and the ATMD with the parameter-independent algorithm can still completely mitigate the VIV of bridges. At the same time, it required smaller maximum control force. Finally, note that CSI plays a role here that is similar a stiffness-like element when the control command is zero, and it affects the control result of the ATMD both in well-tuned case and mis-tuning case. It is recommended that researchers include realistic actuator models in their studies.

ACKNOWLEDGMENTS

This study was financially supported by National Natural Science Foundation of China (52108442), China Postdoctoral Science Foundation (2021M690619), Jiangsu Planned Projects for Postdoctoral Research Funds (2021K099A), National Program on Key R&D Project of China (2020YFB2103500), Key Research and Development Plan of Anhui Province (202104g01020002), and Program of Chang Jiang Scholars of Ministry of Education. These supports are gratefully acknowledged.

CONFLICT OF INTEREST

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

AUTHOR CONTRIBUTIONS

Jun Dai conceptualized the study, designed the methodology, prepared the software, curated the data, conducted the formal analysis and investigation, validated the results, visualized the data, wrote the original draft, and reviewed and edited the manuscript. Zhao-Dong Xu curated the data, acquired the funding, administered the project, supervised the study, and reviewed and edited the manuscript. Shirley J. Dyke conceptualized the study, supervised the study, and reviewed and edited the manuscript. Zheng-Qing Chen reviewed and edited the manuscript.

DATA AVAILABILITY STATEMENT

All of the data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Dai J, Xu Z-D, Dyke SJ. Robust control of vortex-induced vibration in flexible bridges using an active tuned mass damper. *Struct Control Health Monit*. 2022;29(8):e2980. doi:[10.1002/stc.2980](https://doi.org/10.1002/stc.2980)